2pt-Aware Density Estimates

A plan for per-point density weights consistent with Landy–Szalay

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May 1, 2026

Abstract

We outline a pipeline that turns a clustered point catalog (data $\mathcal{D}$) together with its random catalog ($\mathcal{R}$) into per-point density weights $\{w_i\}$ whose pairwise statistics reproduce the Landy–Szalay (LS) two-point correlation function exactly. Part I gives the binned construction. Part II upgrades it to the Storey-Fisher–Hogg (SFH) basis-function continuous-function estimator, which removes bin artifacts and unifies real- and Fourier-space analysis. Part III interprets the resulting weights as samples from a Gaussian process whose covariance kernel is the measured (or modeled) two-point function, and discusses concrete implementation against the `graphgp` library (Vecchia-approximate, JAX-based, scalable to billions of points).

1 Motivation and overview

The standard analysis pipeline of a galaxy redshift survey produces two catalogs: the data catalog $\mathcal{D} = \{x_i\}_{i=1}^{N_D}$ and a random catalog $\mathcal{R} = \{y_\alpha\}_{\alpha=1}^{N_R}$ encoding the survey window. The two-point correlation function is then estimated as

$$\widehat{\xi}_{\rm LS}(r) \;=\; \frac{DD(r) - 2\,DR(r) + RR(r)}{RR(r)}, \tag{1}$$

where DD, DR, RR are the suitably normalized pair counts. Equation (1) is the gold standard estimator: minimum variance for small ξ, robust to the survey geometry through the RR normalization, and immune to the integral constraint at leading order.

What is rarely produced as a primary data product is a per-point density contrast w_i attached to each galaxy. Such a quantity would unlock density-split statistics, environmental analyses, halo bias studies, and field-level inference — all without re-invoking the random catalog. The goal of this document is to specify a concrete construction such that the weights $\{(x_i, w_i)\}$ exactly reproduce $\widehat{\xi}_{\rm LS}$ when subsequently analyzed by simple weighted pair sums of the data alone. In other words: *the random catalog is integrated out into the weights, and downstream analyses become much cleaner and more reusable.*

The construction proceeds in three layers of increasing sophistication, each preserving the previous layer's correctness:

Part I. Binned LS-consistent weights. Run standard binned LS to obtain $\widehat{\xi}_j$ in N_b separation bins. Solve a linear system for $\{w_i\}$ such that weighted DD-only pair counts in each bin reproduce $\widehat{\xi}_j$.

Part II. SFH continuous-function basis. Replace bin indicators with smooth basis functions $f_\alpha(r)$ (cubic splines, BAO templates, or compensated bandpasses). Solve in basis space; the system is smaller, smoother, and natively connects to the power spectrum via Bessel bases.

Part III. Gaussian process interpretation. The weights are samples (or posterior means) of a GP whose covariance kernel is the measured two-point function. Use `graphgp` to scale the construction to $N_D \sim 10^9$ points with linear time and memory.

The unifying observation, drawn out in Section 3.4, is that all three constructions are different views of the same object: the smoothed density field $\widehat{\delta}(x)$ sampled at the data positions, with covariance set by $\xi(r)$.

2 Part I: Binned LS-consistent per-point weights

2.1 Setup

Let the survey have data catalog $\mathcal{D}$ of size N_D and random catalog $\mathcal{R}$ of size N_R. Choose log-spaced separation bins $\{B_j\}_{j=1}^{N_b}$ covering the scientific range; for typical LSS work, 20–30 bins per decade in r from $\sim 0.1\,\mathrm{Mpc}$ to $\sim 200\,\mathrm{Mpc}$ is adequate. Compute the standard normalized pair counts DD_j, DR_j, RR_j with a dual-tree pair finder (e.g. `Corrfunc`). Compute the LS estimator $\widehat{\xi}_j$ in each bin via (1).

2.2 The weight assignment problem

We seek per-data-point weights w_i, with mean adjusted so that $w_i = 1 + \widehat{\delta}_i$ for some local overdensity $\widehat{\delta}_i$, such that

$$\sum_{i<k} w_i\, w_k \mathbb{1}[r_{ik} \in B_j] \;\stackrel{!}{=}\; \left(1 + \widehat{\xi}_j\right) \cdot RR_j \quad \text{for each bin } j. \tag{2}$$

Expanding $w_i w_k = 1 + \widehat{\delta}_i + \widehat{\delta}_k + \widehat{\delta}_i \widehat{\delta}_k$ and using $DD_j = \sum_{i<k} \mathbb{1}[r_{ik} \in B_j]$, equation (2) reduces to

$$\sum_{i<k} \widehat{\delta}_i\, \widehat{\delta}_k \mathbb{1}[r_{ik} \in B_j] \;=\; \widehat{\xi}_j \cdot RR_j \;-\; 2 \sum_i \widehat{\delta}_i\, b_i^{(j)}, \tag{3}$$

where $b_i^{(j)}$ is the number of data points within bin j of point i. Equation (3) is a quadratic system in $\{\widehat{\delta}_i\}$, with N_b equations and N_D unknowns.

2.3 Linearization via the field interpretation

The quadratic system is reformulated linearly by treating $\widehat{\delta}_i$ as the value of an underlying continuous density field $\widehat{\delta}(x)$ at the data point location. With covariance $\mathrm{Cov}[\widehat{\delta}(x), \widehat{\delta}(y)] = \xi(|x-y|)$ assumed equal to $\widehat{\xi}(r)$ from the LS output, the field statement of (2) is linear: the sample covariance of $\{\widehat{\delta}_i\}$ matches the binned target. The unique minimum-variance linear estimator solving this is the Wiener filter,

$$\widehat{\delta}_i \;=\; \left[\, (\mathbf{C} + \mathbf{N})^{-1} \left(\mathbf{n} - \bar{n}\right) \right]_i, \qquad C_{ij} = \widehat{\xi}(r_{ij}), \quad N_{ii} = 1/\bar{n}(x_i), \tag{4}$$

where $\mathbf{n} - \bar{n}$ is the (centered) data indicator at each point and $\bar{n}(x)$ is the local mean density estimated from the random catalog. Equation (4) is the core deterministic construction: once you have run LS, you have $\widehat{\xi}_j$, and the per-point weights are obtained by one linear solve.

2.4 Computational considerations

- For $N_D \lesssim 10^4$, dense Cholesky factorization of $\mathbf{C} + \mathbf{N}$ is direct and well-conditioned.
- For $N_D \sim 10^5$–10^6, conjugate gradient with sparse approximation of $\mathbf{C}$ (truncate to nearest neighbors at each point) is appropriate. Cost $O(N_D^2)$ memory but $O(N_D \log N_D)$ per CG iteration.
- For $N_D \gtrsim 10^7$, Vecchia approximation (Part III) is required.

2.5 Validation

The construction is validated by *re-pairing*: compute the weighted DD pair count

$$\widehat{DD}_w(r) \;=\; \sum_{i<k} w_i w_k \mathbb{1}[r_{ik} \in \mathrm{bin}(r)] \tag{5}$$

on $\{(x_i, w_i)\}$ alone, and verify that $\widehat{DD}_w(r)/RR(r) - 1 \approx \widehat{\xi}_{\mathrm{LS}}(r)$ within the expected statistical noise. The random catalog can then be discarded; $\{(x_i, w_i)\}$ is the new primary data product.

2.6 Limitations of the binned construction

The binned construction has two practical drawbacks that motivate Part II:

1. **Bin-edge artifacts.** The hard indicator $\mathbb{1}[r \in B_j]$ introduces ringing in the recovered $\widehat{\delta}_i$ values. Smoothing via larger bins reduces ringing but loses information.
2. **Bin number tradeoff.** As discussed in Section 2.7, N_b must be chosen by balancing statistical noise per bin against weight-system conditioning. The optimum is ∼25 bins/decade, giving $N_b \sim 75$ for typical surveys — still noisy on adjacent bins.

2.7 Bin number revisited

The standard error on $\widehat{\xi}_j$ scales as $\sqrt{2/N_{\mathrm{pairs},j}}$, where $N_{\mathrm{pairs},j} \propto N_D^2 \cdot \Delta r_j \cdot r_j^2/V$. Going to finer bins makes adjacent bins increasingly correlated and noisy, with no genuine information gain past the point of cosmic-variance saturation. In practice, log-spaced bins with $\Delta r/r \sim 0.1$–0.2 give the best information-per-parameter, but the discontinuous bin structure remains.

This motivates the basis-function approach.

3 Part II: SFH basis-function construction

3.1 The continuous-function estimator

Storey-Fisher & Hogg (2021) replaced bin indicators with smooth basis functions $\{f_\alpha(r)\}_{\alpha=1}^{N_\alpha}$, modelling

$$\xi(r) \;=\; \sum_{\alpha=1}^{N_\alpha} \theta_\alpha \, f_\alpha(r), \tag{6}$$

and projecting the LS pair sums onto the basis. The basis-projected pair counts are

$$DD_\alpha = \sum_{(i,k)\in DD} f_\alpha(r_{ik}), \quad DR_\alpha = \sum_{(i,\beta)\in DR} f_\alpha(r_{i\beta}), \quad RR_\alpha = \sum_{(\alpha,\beta)\in RR} f_\alpha(r_{\alpha\beta}), \tag{7}$$

and the SFH estimator solves a least-squares system for the basis coefficients $\widehat{\theta}_\alpha$ such that $\sum_\alpha \widehat{\theta}_\alpha \int f_\alpha(r) g(r)\, dr$ best matches $(DD_\alpha - 2DR_\alpha + RR_\alpha)/RR_\alpha$ for any test function g. The resulting $\widehat{\xi}(r) = \sum_\alpha \widehat{\theta}_\alpha f_\alpha(r)$ is a continuous estimator with $N_\alpha \ll N_b$ degrees of freedom (typically $N_\alpha \sim 5$–15 cubic splines is adequate where $N_b \sim 75$ bins were needed). The reference implementation is `suave`.

3.2 Weight assignment in basis space

The weight-assignment problem (2) becomes

$$\sum_{i<k} w_i\, w_k\, f_\alpha(r_{ik}) \;\stackrel{!}{=}\; \widehat{\theta}_\alpha \cdot M_\alpha \quad \text{for each basis index } \alpha, \tag{8}$$

with $M_\alpha = \sum_{i<k} f_\alpha(r_{ik})$ a normalization constant computable from the data alone. This is a system of $N_\alpha \sim 10$ equations in N_D unknowns — massively over-determined for any modern catalog, hence well-conditioned for least-squares solution with a smoothness or minimum-variance regularizer.

The basis-form Wiener filter analog of (4) is

$$\widehat{\delta}_i \;=\; \big[\,(\mathbf{C}+\mathbf{N})^{-1}(\mathbf{n}-\bar{n})\,\big]_i, \qquad C_{ij} = \sum_\alpha \widehat{\theta}_\alpha f_\alpha(r_{ij}), \tag{9}$$

which differs from (4) only in how $\widehat{\xi}(r)$ is represented (smooth basis sum versus binned histogram). The basis form is preferred because C_{ij} is now a smooth function of r_{ij} with no bin artifacts.

3.3 Choice of basis

Three natural choices, each with distinct advantages:

Cubic splines. Smooth, locally supported, well-conditioned. Default choice for general two-point function estimation.

Spherical Bessel modes $f_\alpha(r) = j_0(k_\alpha r)$. Each $\widehat{\theta}_\alpha$ is directly an estimate of the power spectrum $\widehat{P}(k_\alpha)$ via the Hankel transform. *The same pair-counting pass yields both $\widehat{\xi}(r)$ and $\widehat{P}(k)$ simultaneously.* The weight assignment in this basis recovers the FKP weighting scheme as the diagonal of the Wiener filter.

Compensated bandpass kernels. Polynomial-compensated functions $f_\alpha(r)$ supported on $[r_{\min,\alpha}, r_{\max,\alpha}]$ with $\int f_\alpha(r) r^2 dr = 0$ realize a wavelet-like decomposition of $\xi(r)$. This basis cleanly handles small-scale modeling failures by placing a hard cutoff at $r_{\min,\alpha}$, while preserving statistical efficiency at the target scale.

For the dyadic compensated-bandpass cascade, the basis functions are the auto-convolutions of the kernel K_ℓ at each level ℓ, and the coefficients $\widehat{\theta}_\ell$ are precisely the sibling covariances $\bar{\xi}_\ell$ that drive the variance cascade $\sigma^2(R_\ell) = \frac{1}{4}\sigma^2(R_{\ell-1}) + \frac{3}{4}\bar{\xi}_{\ell-1}$.

3.4 Unification of real- and Fourier-space analysis

Choosing the Bessel basis makes the unification explicit. The same basis-projected pair-counting pass yields:

1. $\widehat{\xi}(r)$ at any r in the basis range (real space).

2. $\widehat{P}(k)$ at the basis k-values via Bessel transform (Fourier space).

3. Per-point weights w_i via (9).

4. FKP-equivalent mode-by-mode optimal weighting in the appropriate limit.

Practical advantages over standard FFT-based pipelines:

- **High dynamic range.** Pair counting has no grid; the basis functions span as wide a range in r (and equivalently k) as desired, unconstrained by Nyquist or aliasing. A single SFH pass can span 4–5 decades in scale on modern hardware.
- **Complex window handling.** The pair-count form $(DD - 2DR + RR)/RR$ handles arbitrary survey masks and selection functions cleanly through the random catalog, where FFT-based pipelines require careful window-function deconvolution.
- **No mass-assignment biases.** No CIC/TSC corrections, no aliasing near Nyquist.

The cost: $O(N_D \log N_D)$ pair finding scales with N_D, whereas FFT scales with grid size; for very dense surveys at large scales, FFT remains faster. A hybrid pipeline (SFH for small-to-intermediate scales, FFT for largest scales) is the practical optimum.

4 Part III: Gaussian process interpretation and `graphgp`

4.1 The GP interpretation

The Wiener filter form (9) is exactly the conditional mean of a Gaussian process $\widehat{\delta}(x) \sim \mathrm{GP}(0, \xi(\,\cdot\,))$ given Poisson-sampled data on the survey window. More precisely:

- The covariance kernel of the GP is $k(x, y) = \widehat{\xi}(|x - y|)$, the measured (or modeled) two-point function expressed via the SFH basis.
- The data at point x_i is the local overdensity $(n(x_i) - \bar{n}(x_i))/\bar{n}(x_i)$, with measurement noise from finite Poisson sampling.
- The weights w_i are the GP posterior mean evaluated at the data positions, $\widehat{\delta}_i = \mathbb{E}[\widehat{\delta}(x_i) \mid \text{data}]$.
- Sampling new $\widehat{\delta}(x)$ values at unobserved locations uses the standard GP conditional distribution: a single linear algebra operation per query point.

This framing unlocks much more than just per-point weights:

Density at unsampled locations. The GP posterior gives $\widehat{\delta}(x^*)$ with calibrated uncertainty at any x^*, including masked regions (inpainting) and grid points (continuous field reconstruction).

Constrained mock generation. Sample new realizations of the field conditioned on observed galaxy positions, producing constrained mock catalogs that respect both the data and the chosen ξ.

Bayesian model comparison. The GP marginal likelihood $\log p(\text{data} \mid \xi_{\text{model}})$ is computable from the same Cholesky factorization, enabling cosmological model comparison directly at the point-process level.

Higher-order observables. Computing the GP posterior on a grid yields a smooth density field on which any non-Gaussian statistic (bispectrum, density-PDF, peak counts, void counts) can be evaluated.

4.2 The scalability problem

The naive GP regression cost is $O(N_D^3)$ for Cholesky factorization of $\mathbf{C}+\mathbf{N}$ and $O(N_D^2)$ memory for storage. For modern surveys with $N_D \sim 10^7$–10^9, this is intractable: 10^{18}–10^{27} operations and petabytes of memory.

The standard remedy is the *Vecchia approximation*: replace the full conditional density with a product of conditionals on k nearest neighbors,

$$p(\widehat{\delta}_1, \ldots, \widehat{\delta}_{N_D}) \approx \prod_{i=1}^{N_D} p\big(\widehat{\delta}_i \,\big|\, \widehat{\delta}_{N(i)}\big), \tag{10}$$

where $N(i)$ is the set of k nearest preceding neighbors of point i. This reduces cost to $O(N_D k^3)$ time and $O(N_D k)$ memory; for $k \sim 20$–50, this is linear in N_D and practical for billion-point catalogs.

4.3 `graphgp`: the implementation

The `graphgp` package by Dodge & Frank (Stanford ISM group) provides the Vecchia approximation in JAX with optional CUDA acceleration. From the project documentation, `graphgp`:

- Generates GP realizations with approximately stationary, decaying kernels.
- Scales to billions of parameters with linear time and memory requirements.
- Handles arbitrary point distributions with large dynamic range.
- Provides exact inverse and log-determinant via the Vecchia factorization, enabling both sampling and likelihood evaluation.
- Has a JAX implementation with optional custom CUDA extension for Nvidia GPUs.

The key data structure is a `Graph` containing the points, the nearest-neighbor connections for sequential conditioning, and offsets specifying batches that can be generated in parallel. Covariance kernels are passed as discretized $k(r)$ functions on a tabulated r-grid.

4.4 Concrete pipeline using `graphgp`

The full 2pt-aware density estimation pipeline implemented with `graphgp` reads:

1. **Run SFH estimator.** Use `suave` (or a custom implementation) on the data and random catalogs with a chosen basis to obtain $\widehat{\theta}_\alpha$ and the continuous $\widehat{\xi}(r) = \sum_\alpha \widehat{\theta}_\alpha f_\alpha(r)$.
2. **Tabulate the kernel.** Discretize $\widehat{\xi}(r)$ on a logarithmic r-grid spanning the relevant scales (typically 1000 points from $r_{\min}$ to $r_{\max}$, with optional jitter on the diagonal to handle finite-precision degeneracy at small separations). This is the input format expected by `graphgp`.
3. **Build the Vecchia graph.** Call `gp.build_graph(points, n0=100, k=20)` on the data positions to construct the nearest-neighbor graph for sequential conditioning. The parameter `n0` controls the size of the initial dense block; `k` controls the number of neighbors per conditional.
4. **Compute weights.** The per-point density weights $\widehat{\delta}_i$ are the GP posterior mean given the Poisson-sampled data. With `graphgp`, this is one application of the inverse Cholesky factor to the centered data vector $(\mathbf{n} - \bar{n})$. The cost is $O(N_D k)$ time and memory.

5. **Validate.** Compute weighted DD pair sums on $\{(x_i, w_i)\}$ alone and verify recovery of $\widehat{\xi}_{\mathrm{LS}}(r)$ within the basis approximation error.

6. **Discard $\mathcal{R}$.** The random catalog has done its job — its information is now encoded in the per-point weights.

4.5 Worked example (sketch)

A minimal demonstration on a halo catalog of $N_D \sim 10^6$ points:

```
import jax
import graphgp as gp
import numpy as np

# Step 1-2: Run SFH and tabulate the kernel
# (assume xi_r is a callable returning xi at a given r)
r_grid = np.logspace(-1, 2.3, 1000)   # 0.1 to ~200 Mpc
xi_grid = xi_r(r_grid)
covariance = (r_grid, xi_grid)

# Step 3: Build the Vecchia graph
points = halo_positions    # shape (N_D, 3)
graph = gp.build_graph(points, n0=100, k=30)

# Step 4: Compute the per-point weights
# (centered local overdensity from data and randoms)
delta_data = compute_local_overdensity(points, randoms)
weights = gp.apply_inverse_cholesky(graph, covariance, delta_data)

# Step 5: Validate by re-pairing
xi_recovered = weighted_pair_sum(points, weights, r_grid)
assert np.allclose(xi_recovered, xi_grid, rtol=0.05)

# Step 6: Discard randoms; weights are the new primary product
np.savez('halo_catalog_with_weights.npz',
         positions=points, weights=weights)
```

The whole pipeline runs in minutes on a single GPU for $N_D \sim 10^6$ and scales linearly to $N_D \sim 10^9$.

4.6 Connections to `graphgp`'s design choices

Two of `graphgp`'s design decisions are particularly well-aligned with the 2pt-aware density construction:

1. **Discretized covariance interface.** `graphgp` accepts the covariance as a tabulated $(r, k(r))$ tuple, which is exactly the output format of an SFH estimator: $\widehat{\xi}(r)$ on a chosen r-grid. The interface is direct.

2. **Iterative Charted Refinement heritage.** `graphgp` evolved from ICR [6], which itself was designed for multiscale cosmological field reconstruction. The dyadic / cascade structure that

motivates compensated bandpasses in Part II is structurally the same as the multi-resolution graph that ICR/`graphgp` uses internally.

4.7 Beyond density estimation

Once the GP framework is in place, several extensions are natural:

Multi-tracer GPs. Handle multiple galaxy types with cross-covariance kernels $\xi_{ij}(r)$ measured between populations. The weight assignment produces per-population density fields with consistent cross-correlations.

Anisotropic kernels. Encode redshift-space distortions by allowing $\xi(\mathbf{r})$ to depend on the line-of-sight angle. The Vecchia approximation is unchanged; only the kernel evaluation must be updated.

Hierarchical / cascade kernels. The compensated bandpass cascade becomes a sparsity-inducing structure on the GP covariance: each level contributes a separate, scale-localized kernel. The total covariance is a sum of cascade-level kernels, and the Vecchia graph naturally captures the multi-scale neighbor structure.

Field-level inference. The GP marginal likelihood enables direct cosmological-parameter inference at the field level, without summary statistics. The kernel parameters become the inference targets.

5 Summary and recommended next steps

The 2pt-aware density estimation pipeline progresses through three stages, each preserving correctness and extending capability:

1. **Binned LS-consistent weights** (Part I): the conceptually simplest construction. Run standard LS, solve a small linear system, get per-point weights. Useful for proof-of-concept on small catalogs.
2. **SFH basis-function weights** (Part II): the practically preferred construction. Smooth basis avoids bin artifacts; Bessel basis unifies real- and Fourier-space analysis; compensated-bandpass basis enables small-scale-robust cascade analysis.
3. **GP-based weights via `graphgp`** (Part III): the scalable production form. Vecchia approximation makes GP regression feasible for billion-point catalogs. The same machinery enables density reconstruction at unsampled locations, constrained mock generation, Bayesian model comparison, and field-level inference.

The recommended next steps for actual implementation are:

1. Prototype Part I on a small ($N_D \sim 10^4$) halo catalog from a cosmological simulation. Validate that re-pairing recovers the input $\widehat{\xi}(r)$ to within expected statistical noise.
2. Generalize to Part II using `suave` for the SFH estimator. Compare cubic-spline, Bessel, and compensated-bandpass bases for fixed catalog and kernel.
3. Scale to Part III via `graphgp` on the full halo catalog ($N_D \sim 10^7$). Validate weight quality by cross-checking against ground-truth dark matter density at halo locations from the simulation.
4. Apply to real survey data (DESI, Euclid) once validated on simulations. The output — a galaxy catalog with per-galaxy density weights at one or several scales — is the natural successor

data product to the current galaxy + random catalog format, and would substantially simplify downstream analyses.

The unifying principle is that the random catalog has done its work once the weights are computed; downstream science (density-split, environmental analyses, cross-correlations, higher-order statistics, field-level inference) all proceeds from $\{(x_i, w_i)\}$ alone, in a much cleaner and more reusable form.

Acknowledgements. The `graphgp` package is developed by Benjamin Dodge and Philipp Frank in the Stanford ISM group, with guidance from Susan Clark. The Vecchia approximation builds on the long literature in spatial statistics (Vecchia 1988, Datta et al. 2016, Katzfuss & Guinness 2021) and its cosmological-multiresolution variants (Frank et al. 2022, Iterative Charted Refinement).

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